

VISHAVJEET SINGH BHATI

+91 9509600521 bhatiloonkha12@gmail.com linkedin github

Education

Indian Institute of Information Technology, Design and Manufacturing

Integrated B.Tech + M.Tech in Computer Science Engineering; CGPA: 7.01/10

Aug 2024 – May 2029

Kurnool, Andhra Pradesh

Experience

ML Intern – AI/ML (LLM Quantization)

March 2025 – Present

Indian Institute of Technology Jammu

Remote

- Developed a **post-training quantization pipeline** for LLMs applying INT8/INT4 weight quantization, reducing model size by **up to 75%** with under **2% accuracy degradation**.
- Benchmarked quantized models across architectures on **latency, throughput, and memory footprint**; automated **per-layer sensitivity analysis** to identify optimal compression thresholds.

Research Intern – Computer Vision (Welding Defect Detection)

May 2025 – Present

Indian Institute of Technology Mandi

On-site

- Built a **CNN-based welding defect detection system** classifying defects (porosity, cracks, incomplete fusion) on industrial weld imagery with a complete **preprocessing and augmentation pipeline**.
- Integrated the trained model into a **real-time inference pipeline with defect localization**, supporting automated quality control and reducing manual inspection overhead.

Projects

Fractal Intelligence – Market Prediction | Python, Streamlit, Scikit-learn, Plotly, yFinance

[GitHub](#) | [Live App](#)

- Leveraged **Hurst Exponent (R/S analysis)** to analyze **5+ years** of financial time-series data (**1M+ rows**), identifying trending vs. mean-reverting market regimes.
- Developed 20+ volatility and return features to train a **200-tree Random Forest** model, achieving **70–75% backtest accuracy**.
- Constructed a real-time **Streamlit + Plotly dashboard** utilizing data caching to reduce inference latency by **40%**.

Meaning-Preserving AI Enhancer | React, Vite, Groq API (Llama 3), NLP, Cosine Similarity

[GitHub](#) | [Live App](#)

- Engineered an AI writing assistant powered by **Llama 3 (Groq API)** and **React + Vite**, achieving **sub-1s inference latency** for a seamless user experience.
- Integrated **semantic similarity scoring (TF + cosine similarity)** to ensure **90%+ meaning preservation accuracy** during text enhancement.
- Optimized API usage by implementing **3x auto-retry logic** and client-side processing, reducing total API calls by **30%**.

Heart Disease Prediction System | Python, Scikit-learn, Streamlit, Plotly

[GitHub](#) | [Live App](#)

- Developed a Random Forest classifier achieving **85% accuracy**, integrating it into a production-ready pipeline using **Pickle** and **Scikit-learn** for real-time risk assessment.
- Built a high-performance Streamlit dashboard featuring **Plotly-driven gauge charts** and utilized **Resource Caching** to optimize model loading and ensure sub-second response times.

SmartNews Analyzer | Python, Flask, Transformers (BERT/BART), Spacy, NLTK, Scikit-learn

[GitHub](#)

- Developed a hybrid NLP pipeline utilizing **BERT (DistilBERT)** for sentiment analysis and **BART** for abstractive summarization, supporting both deep learning and classical ML (TF-IDF) modes.
- Integrated **spaCy's NER models** to extract and categorize entities (GPE, ORG, PERSON) and implemented ensemble voting logic (VADER, TextBlob, BERT) to increase sentiment prediction reliability.

Technical Skills

Languages: Python, C/C++,

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Matplotlib, Plotly, Streamlit

Machine Learning & NLP: Classification, Regression, Random Forest, Feature Engineering, EDA, TF-IDF, Sentiment Analysis, Text Classification, Time Series Analysis, Model Evaluation (Accuracy, F1, ROC-AUC), SMOTE

Data Engineering & Deployment: Data Cleaning, Web Scraping, REST APIs, Dashboard Development, Model Deployment, Caching/Optimization, Real-time Analytics

Tools & Platforms: Git, GitHub, Linux, VS Code, Jupyter Notebook, yFinance API, Telegram Bot API

Embedded Systems & IoT: ESP32, Arduino IDE, Sensor Integration, I2C/UART, Interrupt-driven Programming

Achievements

Secured a **CodeChef** rating of **1400** and a **CodeForces** rating of **1100**, demonstrating algorithmic proficiency through competitive programming contests.